TS1

Sunday, 22 March 2026 10:59 AM

Ejercicio 2:

Dado h[n] = δ[n] - δ[n - 4], encontrar y[n] = x[n] * h[n] para cada una de las siguientes x[n]:

a) x[n] = cos(ω₀n. T_s). Expresar la respuesta como un único coseno de la forma A cos(ω₀. n . T_s + φ).

b) x[n] = (1/2)^n u[n].

c) x[n] = u[n + 1] - u[n - 2].

Para V: 3 un 1; 2 DR $h(n) = \delta(n) - \delta(n-4)$

$h(0) = \delta(0) - \delta(-4) = 1$

$h(1) = \delta(1) - \delta(-3) = 0$

$h(2) = \delta(2) - \delta(-2) = 0$

$h(3) = \delta(3) - \delta(-1) = 0$

$h(4) = \delta(4) - \delta(0) = -1$

Para $n > 4$ $h(n) = 0$

a) $x(n) = \cos(\omega_0 n T_s)$

$y(n) = x(n) * h(n)$

$h(n) = \delta(n) - \delta(n-4)$

$\rightarrow y(n) = x(n) * (\delta(n) - \delta(n-4))$

$= \sum_{-\infty}^{\infty} x(i) \cdot h(n-i)$

$y(n) = \sum_{-\infty}^{\infty} x(i) \cdot (\delta(n-i) - \delta(n-i-4))$

Sabemos que $\delta(0) = 1$ y $\delta(\neq 0) = 0$, Para que valuar en i $\delta(i) = 1$?

$n - i = 0 \leftrightarrow \boxed{i = 0}$

$n - i - 4 = 0 \leftrightarrow \boxed{i = -4}$

$y(n) = x(n) - x(n-4)$

$$\boxed{y(n) = \cos(\omega_0 n T_s) - \cos(\omega_0 (n-4) T_s)}$$

Tengo que expresarlo como $A \cos(\omega_0 n T_s + \varphi)$

Prof transformo que T.T.G.

$$\boxed{\cos(A) - \cos(B) = -2 \operatorname{sen}\left(\frac{A+B}{2}\right) \cdot \operatorname{sen}\left(\frac{A-B}{2}\right)}$$

$A = \omega_0 n T_s$, $B = \omega_0 (n-4) T_s$

$y(n) = \cos(A) - \cos(B) = -2 \operatorname{sen}\left(\frac{A+B}{2}\right) \cdot \operatorname{sen}\left(\frac{A-B}{2}\right)$

$y(n) = -2 \operatorname{sen}\left(\frac{\omega_0 n T_s + \omega_0 (n-4) T_s}{2}\right) \operatorname{sen}\left(\frac{\omega_0 n T_s - \omega_0 (n-4) T_s}{2}\right)$

$y(n) = -2 \operatorname{sen}\left(\frac{\omega_0 n T_s + \omega_0 n T_s - \omega_0 4 T_s}{2}\right) \cdot \operatorname{sen}\left(\frac{\omega_0 n T_s - \omega_0 n T_s + \omega_0 4 T_s}{2}\right)$

$y(n) = \underbrace{-2}_{A} \operatorname{sen}(\omega_0 n T_s - 2\omega_0 T_s) \underbrace{\operatorname{sen}(2\omega_0 T_s)}_{A}$

$$\boxed{\operatorname{sen}(\gamma) = \cos\left(\gamma - \frac{\pi}{2}\right)}$$ Prof.

$\operatorname{sen}(\omega_0 (n-2) T_s) = \cos\left(\omega_0 (n-2) T_s - \frac{\pi}{2}\right)$

$y(n) = -2 \operatorname{sen}(2\omega_0 T_s) \cdot \cos\left(\omega_0 (n-2) T_s - \frac{\pi}{2}\right)$

Puedo pasar ω^{-1} $\rightarrow$ $-\frac{\pi}{2} \rightarrow$ suma π

$$\boxed{y(n) = 2 \operatorname{sen}(2\omega_0 T_s) \cdot \cos\left(\omega_0 (n-2) T_s + \frac{\pi}{2}\right)}$$ Rta

b) $x(n) = \left(\frac{1}{2}\right)^n u(n)$

$y(n) = \sum_{-\infty}^{\infty} x(i) \cdot (\delta(n-i) - \delta(n-i-4))$

$$\boxed{y(n) = \frac{1}{2^n} u(n) - \left(\frac{1}{2}\right)^{n-4} u(n-4)}$$

$n < 0 \rightarrow y(n) = 0$

$0 \leq n \leq 3 \rightarrow y(n) = \frac{1}{2^n} u(n)$

$n \geq 4 \rightarrow y(n) = \frac{1}{2^n} u(n) - \left(\frac{1}{2}\right)^{n-4} u(n-4)$

c) $x(n) = u(n+1) - u(n-2)$

$h(n)$

Primero veamos como es $x(n)$

$x(n) = u(n+1) - u(n-2)$

es una "ventana" de -1 a 1

$$x(n) = \begin{cases} 0 & n < -1 \\ 1 & -1 \leq n \leq 1 \\ 0 & n \geq 2 \end{cases}$$

$h(n) = \delta(n) - \delta(n-4)$

$y(n) = x(n) * h(n) = x(n) - x(n-4)$

$= u(n+1) - u(n-2) - (u(n-4+1) - u(n-4-2))$

$$\boxed{y(n) = u(n+1) - u(n-2) - u(n-3) + u(n-6)}$$

Para $n < 1 \rightarrow y(n) = 0$

$x(n) = 0$, $x(n-4) = 0$

Para $-1 \leq n \leq 1$

$x(n) = 1$, Para $x(n-4) = 0$

$x(-3) = x(-4) = x(-3) = 0$

$\rightarrow y(n) = 1$

Para $n \geq 2$

$x(n) = 0$, Pero $x(n-4)$ si $n-4 \in [-1, 1] = 1$

Si $n = 3 \rightarrow y(3) = x(3) - x(3-4) = 0 - 1 = -1$

$n = 4 \rightarrow y(4) = x(4) - x(4-4) = 0 - 1 = -1$

$n = 5 \rightarrow y(5) = x(5) - x(5-4) = 0 - 1 = -1$

$n = 6 \rightarrow y(6) = x(6) - x(6-4) = 0 - 0 = 0$

Antes de $n \geq 6 \rightarrow y(n) = 0$

$3 \leq n \leq 5 \rightarrow \boxed{y(n) = -1}$

Finalmente

$$y(n) \begin{cases} 0 & n \leq -2 \\ 1 & -1 \leq n \leq 1 \\ 0 & n = 2 \\ -1 & 3 \leq n \leq 5 \\ 0 & n \geq 6 \end{cases}$$